

An Efficient Deep Learning Based Approach for Toxic and Edible Mushroom Classification

Abstract—Mushrooms have become a highly profitable global industry, significantly contributing to GDP growth in recent days. However, the rising demand for mushrooms increases the risk of consuming toxic or poisonous varieties, which can cause serious health issues. Advancements in Artificial Intelligence and Machine Learning algorithms have significantly improved the ability to classify edible and toxic mushrooms. In this study, we developed a customized deep learning model that integrates the feature extraction capabilities of EfficientNet-B0 with a Support Vector Machine (SVM) for classification. We have implemented several existing deep learning architectures and compared their performance with our proposed model to validate the effectiveness. Our custom model has showed a significant improvement over the traditional and existing deep learning models. It has achieved an accuracy of 99.5% outperforming most of the existing models.

Index Terms—Mushrooms, Toxic Mushrooms, Mushroom Classification, EfficientNet, SVM.

I. INTRODUCTION

Mushrooms are well known as a fancy food item used in several dishes from pizza to pasta, enhancing and enriching their flavors. On the other hand, mushrooms are also recognized for their medical benefits around various parts of the globe. Among the identified edible mushroom species around 700 species are being considered to have medicinal properties Jo et al. (2014). It contains nutrients, antioxidants, anti-inflammatory properties, anti-cancer properties, and some unique compounds that help to boost the immune system, lower cholesterol levels, regulate blood sugars, and so on. Therefore, several diseases that human beings often face can be treated with the particles extracted from edible mushrooms El-Ramady et al. (2022). They are one of the healthiest foods that are grown without any effort

As the demand for mushrooms has increased worldwide, a recent statistical analysis is made where it is said that the global production of mushrooms grew from 6.90 to 10.24 million metric tons in the span of 9 years from 2008 to 2019 and based on the latest statistics it is estimated that the global market size of mushroom is valued at USD 50.3 billion in 2021 and is expected to expand to 136 billion within 2032 according to Market.us. The agricultural sector of Bangladesh has been one of the main economic driving forces as it contributes approximately 20% of the national GDP Zahan et al. (2021). In recent years Bangladesh has also taken an interest in mushroom production for national use and exporting them as it is one of the suitable countries for cultivating mushrooms due to its favorable climates with low production cost and the availability to grow substrates Ferdousi et al. (2020).

Eating mushrooms without properly identifying and confirming their species' edibility is not advised. Less than 1% of all mushrooms are known to be toxic, making them dangerous and occasionally lethal species. The ability to distinguish edible mushrooms from inedible and toxic ones is crucial. This work uses computer vision and deep learning to classify mushrooms into edible, inedible, and dangerous categories. Deep learning is an artificial intelligence subfield of machine learning that mimics how the human brain processes information and creates patterns. Zahan et al. (2021).

Bangladesh produces over 40,000 tons of mushrooms every year and its estimated market value is almost around 70 million USD. Among the mushrooms Oysters and Milky mushrooms are being produced in large amounts on the country's favorable environment and temperature. As the demand for mushrooms has increased, people have less knowledge about various kinds of mushrooms and are endangering themselves by consuming poisonous mushrooms. In a study done by the National Library of Medicine in China, their result confirms that around 10,036 outbreaks were recorded among which 38,676 illnesses, 21,967 hospitalizations, and 788 deaths were confirmed between 2010-2020 Li et al. (2021). Although mushrooms are an important natural resource with nutritional and therapeutic properties, misidentifying toxic species can negatively affect one's health, even death. In Bangladesh, where mushroom cultivation is limited, trustworthy techniques to differentiate between edible or dangerous types are desperately needed. Conventional identification techniques depend on specialized knowledge, which the general public may not always have access to. To close this gap, this project uses deep learning to produce a mushroom categorization tool that is accurate, scalable, and efficient.

The contribution of the performed research is summarized below:

- Improved deep learning model to classify toxic mushrooms could significantly reduce the possibility of marketing and consuming toxic mushrooms.
- Application of customized deep learning architecture in order to earlier detection of toxic mushrooms from images that outperform the state-of-the-architectures. The model is evaluated based on numerous performance metrics.
- This study will pave the way for advancing AI-driven agricultural support at an early stage. Additionally, the findings of this experiment can contribute to more effective mushroom cultivation strategies. Overall, the proposed architecture has the potential to enhance the agri-

cultural sector significantly.

II. BACKGROUND STUDY

Machine Learning is the field of study that enables computers to learn from data. Deep learning is a part of machine learning where AI is used for supervised, unsupervised as well as reinforcement machine learning. A collection of data is used to train the machine so that it can later sort and identify objects according to their classes easily which is often time-consuming for a human to do so. Deep learning uses various types of algorithms such as CNN, ANN, GANs, GNN, KNN, and so on. Depending on the field that is being studied, the appropriate algorithms are used. To conduct research, we need to analyze various types of algorithms and then only we can get the output with accuracy and precision Aravindhaa et al. (2024).

Convolutional Neural Networks are a part of machine learning that is used to distinguish between data with the use of image recognition and it is widely used in the agriculture field for many purposes Devika & Karegowda (2021), Shehab et al. (2023). CNN resulted in significant accuracy and became a standard method in image classification tasks for machine learning after 2012 ILSVRC Wang et al. (2022). Instead of using manually constructed image attributes as classifiers in traditional machine-learning approaches, CNNs may automatically learn the distinguishing features of images/objects from pixel arrangements in images Demirel & Demirel (2023). At the current time, there are two main types of target recognition algorithms. One is a two-stage algorithm such as R-CNN and the other one is a regression-based first-order algorithm such as YOLO. Yolov5 consists of four different models that include Yolo 5s, Yolo V5m, Yolo V5l, and Yolo 5x. They are pre-trained with an 80-class MSCOCO dataset Cengil & ÇâĖnar (2021).

The authors in Demirel & Demirel (2023) have used about 5500 pictures of mushrooms to test train and validate their model. They have applied CNN-Model as well with the help of "Mobilenetv2 GAP flatten fc" which was improvised from "Mobilenet-V2" as it is small and efficient and works with very low memory and processing capacity and it performed with an accuracy of 97.89%. It can be seen being used in leaf disease detection, detection of fruits, and so on. Moysiadis et al. (2023) have claimed that YOLOv5 is one of the fastest object detection algorithms and their result shows that it can detect and classify mushrooms with an F1-score of up to 76.5% and an accuracy of 70%. Wang and others in Wang et al. (2022) also conducted their work using YOLOv5 where they mentioned that it fails sometimes to detect the overlapping or small targets when recognizing mushrooms in experiments. So they had to improve the model so that it would be suitable for the targets which resulted in better performance with up to 99.24

Decision Tree is another supervised machine learning algorithm that is used for classification and regression. It consists of a root node, branches, internal nodes, and leaf nodes, it is

arranged hierarchically where each leaf node represents a decision among multiple choices and the branch node represents a decision. It learns to approximate a sine curve by using the if-then-else rule Tutuncu et al. (2022). Analyzing the research conducted by Shehab et al. (2023), they used 23 species of mushroom and after data cleaning and preprocessing and applying various algorithms, it is concluded that the decision tree outperforms the other algorithms with an accuracy of 98% and precision of 97.14%.

On the other hand, Ottom and others in Ottom et al. (2019) also conducted similar experiments with different datasets. They compared Neural Networks, Decision Tree, and KNN. From the collected dataset they extracted the Eigen histogram and parametric features before applying the algorithms to get better results. In another experiment done by TUTUNCU and others in Tutuncu et al. (2022), more than 8000 images were collected and 22 features were extracted from them to apply the decision tree, Naive Bayes, SVM, and AB where the Decision tree had an accuracy of 98.82% and precision of 100

III. METHODOLOGY

In our study, we have developed a customized deep learning architecture for the early detection of toxic mushrooms. The overall outline of the overall methodology is illustrated in Figure 1.

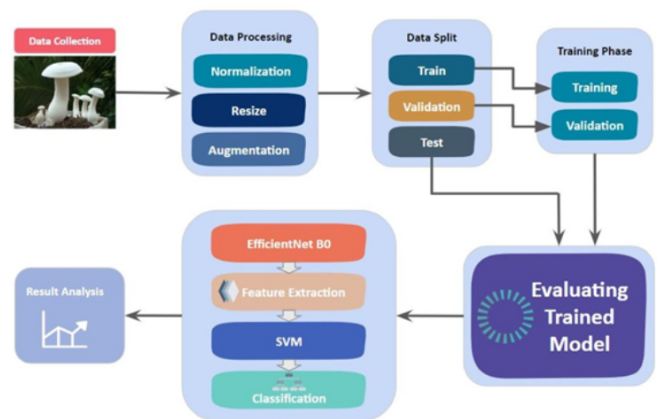


Fig. 1: Methodology of the Research

The process begins with data collection. After acquiring the mushroom images, we have applied several image pre-processing techniques, including normalization, resizing, and augmentation. We then split the dataset into train and test segments following an 8:2 ratio. The training set has been used for iterative training purposes, while the testing set has been kept reserved for evaluation purposes.

Following these steps, we have deployed our customized deep learning architecture alongside several existing deep learning architectures. Each model was trained independently, and individual tests were conducted using the test set for each model. Finally, We assessed the performance and effectiveness of our proposed model and compared it with the other deep learning model to evaluate its superiority.

A. Description of the Dataset

We have collected different types of mushroom image data from the Mushroom Development Institute (Savar, Dhaka). However, the data we collected was not sufficient enough so we have decided to collect data from open-source (iNaturalist website) as well. There were 6 classes of Mushrooms Agaricus Edible, Agaricus Bisporus Edible, Calocybe Indica Edible, Oyster Edible, Amanita Toxic, and Deathcap Toxic. Figure 2 represents the samples from the dataset and the Pie Chart illustrated in Figure 3 shows the class wise distribution of the utilized dataset.

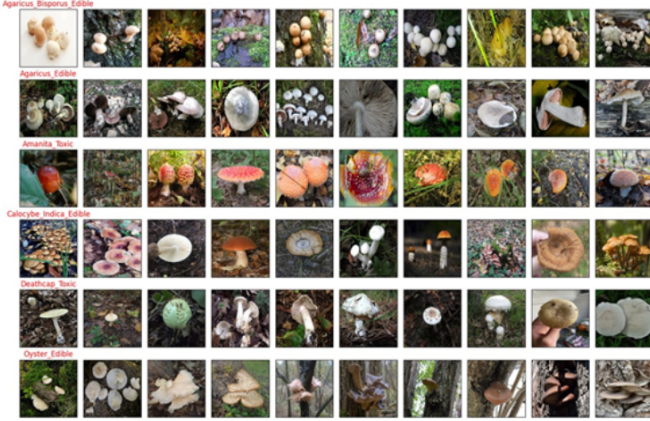


Fig. 2: Sample images from the dataset

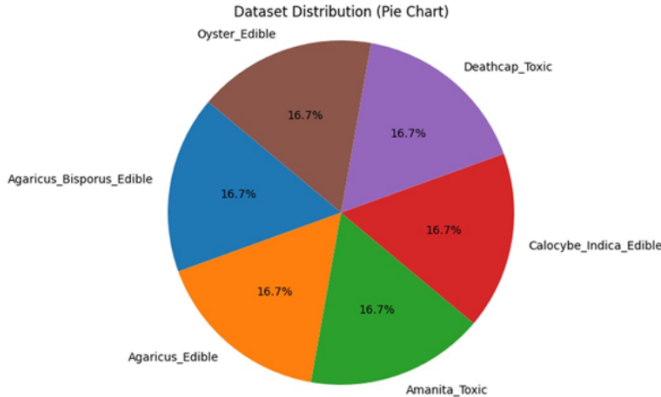


Fig. 3: Sample images from the dataset

B. Proposed Model

In this study, we propose a hybrid deep-learning model for differentiating between edible and toxic mushrooms from mushroom images. The hybrid model is concise of two state-of-the-art models, EfficientNet B0 and Support Vector Machine. EfficientNetB0 is a convolutional neural network architecture that uses a sequence of convolutional layers to interpret input images in the context of feature extraction, capturing hierarchical information at various sizes, ranging

from basic edges to intricate patterns. Compared to conventional CNNs, EfficientNetB0 can extract rich and detailed features with fewer parameters thanks to its compound scaling technique, which uniformly scales the network's depth, width, and resolution. In our proposed model, The most important features of the classification have been extracted from the images using EfficientNetB0.

On the other hand, Support Vector Machine (SVM) is a supervised machine learning algorithm widely used for classification tasks, particularly in scenarios with clear margins of separation between classes. SVM is a well-liked option in machine learning because of its resistance to overfitting, particularly in high-dimensional spaces. Thus, we have utilized SVM for classification purposes. Figure 4 provides an architectural overview of the proposed model.

IV. RESULT ANALYSIS

As mentioned in section III, the training set has been split into 8:2 ratio for training and validation. We have performed 50 iterations of training utilizing 80% of the training data and performed validation after completion of each iteration using the rest of the 20% data. In each iteration, we have observed the accuracy and loss. For the graphs mentioned in Figure 5, we can see that the training accuracy has reached a stable position (close to 1) after the 30th epoch and the validation shows an incremental trend up to 49th epoch. Similarly, the loss has shown a decremental trend for both training and validation.

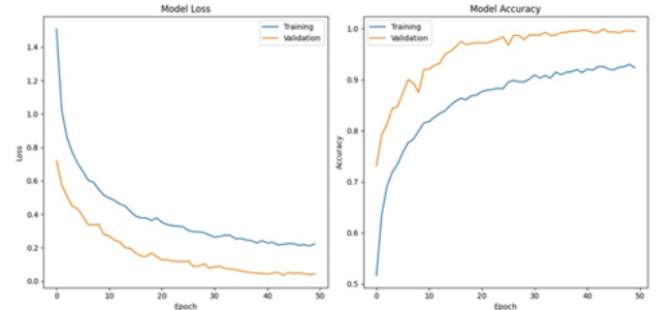


Fig. 5: Learning curves of the proposed model

From the classification report mentioned in Table I, we can see a significant results in each class of the mushrooms.

TABLE I: Classification Report of the proposed model

Class	Precision	Recall	F1-Score
Agaricus_Bisporus_Edible	0.9950	0.9950	0.9950
Agaricus_Edible	0.9945	0.9900	0.9925
Amanita_Toxic	1.0000	1.0000	1.0000
Calocybe_Indica_Edible	0.9901	1.0000	0.9950
Deathcap_Toxic	0.9950	0.9950	0.9925
Oyster_Edible	1.0000	0.9900	0.9950
Accuracy		0.9950	
Macro Avg	0.9950	0.9950	0.9950
Weighted Avg	0.9950	0.9950	0.9950

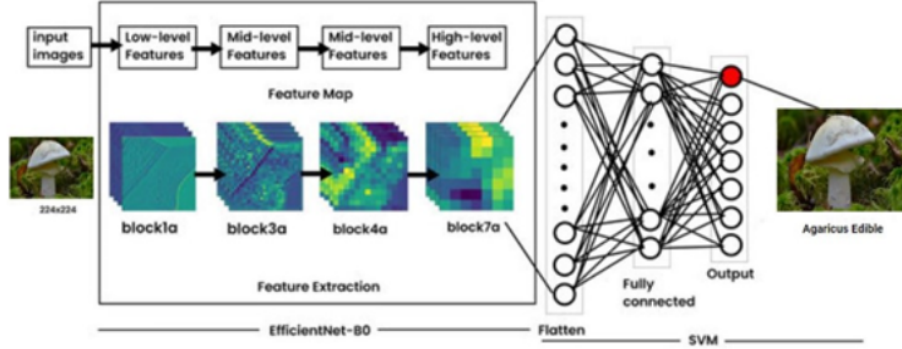


Fig. 4: Architecture of the proposed model

We can see from Table II that the proposed hybrid model has outperformed the other models in terms of accuracy.

TABLE II: Performance comparison based on Accuracy Score

Model	Accuracy
CNN	86.00%
VGG 16	87.93%
ResNet 50	91.37%
Hybrid	99.50%

V. CONCLUSION

This study successfully demonstrates the potential of leveraging advanced machine learning techniques, including a customized hybrid model, for the classification of edible and toxic mushrooms. By achieving superior accuracy, the proposed model underscores its effectiveness and applicability in addressing a critical challenge in mushroom identification. While the current dataset is geographically limited and environmental factors may influence performance, the proposed approach provides a solid foundation for future developments. With further enhancements, such as expanding the dataset, incorporating explainable AI (XAI), and exploring alternative architectures, this work can be extended to real-world applications, such as mobile tools for mushroom classification. Ultimately, the findings of this study highlight the promising role of AI in agriculture and safety, paving the way for innovative solutions in the field.

VI. REFERENCES

- [1] W. S. Jo, M. A. Hossain, and S. C. Park, "Toxicological profiles of poisonous, edible, and medicinal mushrooms", 2014, Korean Society of Mycology. doi: 10.5941/MYCO.2014.42.3.215.
- [2] H. El-Ramady et al., "Edible Mushrooms for Sustainable and Healthy Human Food: Nutritional and Medicinal Attributes," May 01, 2022, MDPI. doi: 10.3390/su14094941.
- [3] A. Gowda Karegowda, "Identification of Edible and Non-Edible Mushroom Through Convolution Neural Network," 2021. [Online]. Available: <http://www.mushroom.world>
- [4] "Mushroom Market Size, Share — CAGR of 9.5%." Accessed: Aug. 08, 2024. [Online]. Available: <https://market.us/report/mushroom-market/>
- [5] N. Zahan and Z. Hasan, "A DEEP LEARNING-BASED APPROACH FOR EDIBLE, INEDIBLE AND POISONOUS MUSHROOM CLASSIFICATION," 2021.
- [6] J. Ferdousi, Z. Al Riyadh, Md. I. Hossain, S. R. Saha, and M. Zakaria, "Mushroom Production Benefits, Status, Challenges and Opportunities in Bangladesh: A Review," *Annu Res Rev Biol*, pp. 1â13, Mar. 2020, doi: 10.9734/arrb/2019/v34i630169.
- [7] "Potential of mushroom farming remains capped — The Business Standard." Accessed: Aug. 08, 2024. [Online]. Available: <https://www.tbsnews.net/economy/agriculture/potential-mushroom-farming-remains-capped-176617>
- [8] W. Li et al., "Mushroom Poisoning Outbreaks in China, 2010â2020," *China CDC Wkly*, vol. 3, no. 24, pp. 518â522, 2021, doi: 10.46234/CCDCW2021.134.
- [9] "MUSHROOM CLASSIFICATION USING MACHINE LEARNING," *International Research Journal of Modernization in Engineering Technology and Science*, May 2023, doi: 10.56726/irjmets38699.
- [10] R. N. Said, A. Sallam, E. A. Shehab, and S. A. Shehab, "The classification of mushroom using ML," Online, 2023. [Online]. Available: <https://kjis.journals.ekb.eg/>
- [11] Y. Wang, L. Yang, H. Chen, A. Hussain, C. Ma, and M. Al-Gabri, "Mushroom-YOLO: A deep learning algorithm for mushroom growth recognition based on improved YOLOv5 in agriculture 4.0," in *IEEE International Conference on Industrial Informatics (INDIN)*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 239â244. doi: 10.1109/INDIN51773.2022.9976155.
- [12] Y. DEMÄ°REL and G. DEMÄ°REL, "Deep Learning Based Approach for Classification of Mushrooms," *Gazi University Journal of Science Part A: Engineering and Innovation*, vol. 10, no. 4, pp. 487â498, Dec. 2023, doi: 10.54287/guj-sa.1355751.
- [13] E. Cengil and A. Äinar, "Poisonous Mushroom Detection using YOLOV5," 2021.

[14] V. Moysiadis, G. Kokkonis, S. Bibi, I. Moscholios, N. Maropoulos, and P. Sarigiannidis, "Monitoring Mushroom Growth with Machine Learning," *Agriculture (Switzerland)*, vol. 13, no. 1, Jan. 2023, doi: 10.3390/agriculture13010223.

[15] K. Tutuncu, I. Cinar, R. Kursun, and M. Koklu, "Edible and Poisonous Mushrooms Classification by Machine Learning Algorithms," in 2022 11th Mediterranean Conference on Embedded Computing, MECO 2022, Institute of Electrical and Electronics Engineers Inc., 2022. doi: 10.1109/MECO55406.2022.9797212.

[16] M. A. Ottom, N. A. Alawad, and K. M. O. Nahar, "Classification of mushroom fungi using machine learning techniques," *International Journal of Advanced Trends in Computer Science and Engineering*, vol. 8, no. 5, pp. 2378â2385, Sep. 2019, doi: 10.30534/ijatcse/2019/78852019.

[17] A. Gowda Karegowda, "Identification of Edible and Non-Edible Mushroom Through Convolution Neural Network," 2021. [Online]. Available: <http://www.mushroom.world>

[18] W. Ketwongsa, S. Boonlue, and U. Kokaew, "A New Deep Learning Model for The Classification of Poisonous and Edible Mushrooms Based on Improved AlexNet Convolutional Neural Network," *Applied Sciences (Switzerland)*, vol. 12, no. 7, Apr. 2022, doi: 10.3390/app12073409.

[19] D. Rianasari, M. Noor Triana, M. R. Dewi, Y. Astutik, and R. Wirawan5, "The Classification of Mushroom Types Using Naive Bayes and Principal Component Analysis," *Jurnal Informatika dan Sains* e-ISSN, vol. 05, no. 02, pp. 2614â8404, 2022.